File: `src/config/seriesCatalog.js`

Category: Metric Registry / Configuration Layer
Runtime: Browser (static configuration + pure helpers)
Primary consumers:

- `useChartPanels.js`
- `LiveReadout.jsx`
- `MinMaxReadout.jsx`
- `LineGraph.jsx`

Overview

Defines the **central registry of all measurable metrics** in the application.

This file acts as the **single source of truth** for:

- Available metrics
- Labels and units
- Default chart colors
- Decimal formatting
- Chart eligibility
- Value extraction logic

All UI and chart logic references this registry instead of directly accessing nested data structures.

This enforces consistency and separation of concerns.

Exported Objects and Functions

`SERIES_CATALOG`

Type

```
Record<string, MetricDefinition>
```

MetricDefinition Structure

```
{
  label: string
```

```
  unit: string
  group: string
  defaultColor: string
  decimals: number
  chartable: boolean
  valueAccessor: (dp: ParsedDataPoint) → number | null
}
```

Purpose

Each key represents a measurable metric (e.g., `voltage_a`, `current_total`, `p_active`).

The registry defines:

- How the metric is displayed (label, unit)
- How it is grouped in UI tables
- Whether it is available in charts
- How its value is extracted from parsed data

Key Field: `valueAccessor`

Each metric defines a function:

```
valueAccessor: (dp) => dp.voltages.a
```

This abstracts data extraction away from the UI.

Charts and tables never access:

```
dp.voltages.a
```

Instead they call:

```
metric.valueAccessor(dp)
```

This design allows:

- Internal data structure changes without modifying UI logic
- Centralized metric definitions
- Consistent access patterns

Metric Grouping

Metrics are categorized by:

- `"Temperature"`
- `"Current"`
- `"Voltage"`
- `"Power"`
- `"Energy"`

Grouping is used for:

- Table organization in LiveReadout
- Section spacing in MinMaxReadout

`ALL_METRIC_KEYS`

Signature

```
ALL_METRIC_KEYS: string[]
```

Returns all keys defined in `SERIES_CATALOG`.

Used by:

- Readout components
- Statistics calculations

`CHART_KEYS`

Signature

```
CHART_KEYS: string[]
```

Derived from `ALL_METRIC_KEYS` where:

```
metric.chartable !== false
```

Defines which metrics can appear in chart dropdown menus.

Prevents:

- Energy totals (non-instantaneous values) from being charted

- Metrics marked as non-visual

`ALL_SERIES_KEYS`

Alias for `CHART_KEYS`.

Maintained for backward compatibility with earlier versions (IMP-Dashboard).

`getGroupedMetricKeys(keys?)`

Signature

```
getGroupedMetricKeys(keys?: string[]) → Array<[groupName: string, keys:
string[]]>
```

Parameters

- Optional array of metric keys.
- Defaults to `ALL_METRIC_KEYS`.

Returns

- An array of `[groupName, keys[]]` pairs.

Behavior

1. Iterates over provided keys.
2. Groups them by their `group` property.
3. Returns grouped entries.

Used by

- `LiveReadout.jsx`
- `MinMaxReadout.jsx`

Side effects

- None (pure function).

Design Rationale

This file centralizes metric configuration to:

- Avoid duplicated labels or units across components
- Maintain consistent formatting
- Simplify adding or removing metrics
- Enforce controlled chart eligibility

It functions as a declarative configuration layer rather than executable logic.

Failure Modes / Edge Cases

- If `valueAccessor` references a missing field, it returns `null` (safe).
- If a key is missing from `SERIES_CATALOG`, dependent components will skip rendering that metric.

Architectural Role

This module forms the bridge between:

```
parseShellyData → structured data
seriesCatalog → semantic meaning of data
UI components → presentation and visualization
```

It prevents UI components from depending on device JSON structure.